

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591 -> 2.365667147	delta=+0.0140696
ratio=1.005983		
CoG x [m]	-0.002024708562 -> -0.002831076877	delta=-0.000806368
ratio=-		
CoG y [m]	-3.052657894e-05 -> 0.000850872954	delta=+0.0008814
ratio=-		
CoG z [m]	0.009509749599 -> 0.01088617665	delta=+0.00137643
ratio=-		
rotor effectiveness 1	1 -> 0.9629222039	delta=-0.0370778 ratio=0.9629222
rotor effectiveness 2	1 -> 0.9749871003	delta=-0.0250129 ratio=0.9749871
rotor effectiveness 3	1 -> 0.9620282105	delta=-0.0379718 ratio=0.96202821
rotor effectiveness 4	1 -> 0.9517498315	delta=-0.0482502 ratio=0.95174983
rotor command lag [s]	0.004997730255 -> 0.2	delta=+0.195002 ratio=-

Inertia matrix [kg m^2]

reference:

```
[ 0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[ 1.901508003e-05, 5.9167305e-08, 0.1289921107]
```

estimated:

```
[ 0.0650136254, -2.281053084e-07, 2.409351998e-05]
[-2.281053084e-07, 0.06529447317, 1.194196988e-05]
[ 2.409351998e-05, 1.194196988e-05, 0.1289945608]
```

Principal moments: reference -> estimated

principal moment 1 [kg m^2]	0.06495264516 -> 0.06501361614	delta=+6.0971e-05
ratio=1.0009387		
principal moment 2 [kg m^2]	0.06500006701 -> 0.06529447112	delta=+0.000294404
ratio=1.0045293		
principal moment 3 [kg m^2]	0.1289921163 -> 0.1289945721	delta=+2.45576e-06
ratio=1.000019		

Optimization

joint dynamics loss: 1.15101493643
Gaussian-prior cost: 0.347059818089
selected strict-ZOH rotor lag [s]: 0.2

Bag failure_1_single

reconstruction position RMSE [m]: 0.00455405798905
reconstruction position component RMSE [m]: [0.003382 0.001131 0.002832]
reconstruction orientation RMSE [deg]: 1.63617908635
gyro consistency RMSE xyz [rad/s]: [0.074699 0.140861 0.025946]
specific-force consistency RMSE xyz [m/s^2]: [0.852605 0.148387 0.276523]
inferred external force RMS xyz [N]: [1.089607 0.394447 3.014231]
inferred external torque RMS xyz [N m]: [0.076188 0.254344 0.088915]
bag dynamics loss: 1.15101493643
gimbal command lag [s] 0.004998683929 -> 0.03015494496

Recorded command timestamp intervals (data-derived)

Bag failure_1_single

rotor: median dt=0.00499773026 s (200.091 Hz), min=2.00271606e-05 s, max=0.0138893127 s
gimbal: median dt=0.00499868393 s (200.053 Hz), min=4.529953e-06 s, max=2.48282003 s

Selected optimizer diagnostics

status=1 success=True nfev=4 njev=4
message='gtol' termination condition is satisfied.
reported optimality=7.994731454630255e-08
initial cost=1.49807485555288
final cost=1.4980747545219253
cost reduction=1.0103095471336587e-07
coordinate step L2=5.622224279402233e-05
final gradient inf-norm=7.994731454630255e-08
gradient tolerance satisfied=True
final Jacobian rank=14
final Jacobian cond(nonzero)=392.2063177156247
max directional finite-difference Jacobian relative error=8.60942559528201e-12